Quantitative Localization and Finite Heat-Contact Certificates for Stationary Integral Varifolds

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Technical Report Assisted by AI Models
www.integralresearch.net
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Technical Report TR-Ram-OpenAI-722-M663

October 9, 2026

Abstract

We study the heat-potential interface in the supplied manuscript *Almost-everywhere regularity of stationary integral varifolds*. For its signed source $f = wL' - \eta(\tau + d\,\mathrm{d}y)$, a direct periodized-Gaussian estimate enlarges the sufficient localization window from $c\eta^2 R/T^2$ to $c\eta R[1 + C_1 T + m\log(1/\eta)]^{-1/2}$. The retained Lipschitz factor is essential to the improved logarithmic budget. An atomic example gives the same scale order as an obstruction under the abstract measure hypotheses, without purporting to be a stationary-varifold example. We also prove explicit positive-bottom Fourier-tail and finite-payoff certificates for the source contact construction. Cumulative contact error, initial heat-source error, and weak smoothing bias have different estimates and cannot be interchanged. The single-scale contact problem has a concave energy; variable-scale contacts need not have a symmetric interaction matrix. The signed-excess induction, center construction, collar estimates, and final regularity theorem remain source inputs, not consequences certified by our finite calculations. Contribution assessment: C3/5, a source-specific enlarged localization window with matching abstract scale obstruction and a domain-correct finite contact error certificate is proved, without independent certification of global regularity.

Contents

1 The geometric problem and the proof boundary **2**
1.1 Architecture of the assigned manuscript 2

2 A larger localization window **3**
2.1 An abstract sharpness boundary 4

3 Zero-mode heat potentials and their finite tails **5**
3.1 A complete tail fixture 6

4 A finite contact certificate with separate initial control **6**

5 Selection geometry, smoothing, and what is not certified **8**

6 Ram comparison, consequences, and limitations **9**
6.1 The original private proof connection 9
6.2 Advantages and costs 9
6.3 Precise downstream boundary 10
6.4 Feasible next steps 10

7 Current primary literature and evidence **10**
7.1 Public status as checked on October 9, 2026 10
7.2 Literal scope and reproducibility 11

8 Conclusion **11**

1 The geometric problem and the proof boundary

Let $m, n \geq 1$, let $U \subset \mathbb{R}^{m+n}$ be open, and let

$$V = \theta\, \mathcal{H}^m \restriction E, \qquad \mu = \|V\|, \qquad \int \operatorname{tr}(P(x) D\mathcal{X}(x))\, \mathrm{d}\mu(x) = 0 \quad (\mathcal{X} \in C^1_c(U; \mathbb{R}^{m+n})).$$

Here E is countably m-rectifiable, θ is positive integer-valued almost everywhere, and $P(x)$ is the orthogonal projection onto the approximate tangent plane. Local finite mass is understood. A regular point has a neighborhood on which $V = q|\Sigma|$ for a smooth embedded minimal m-submanifold and one positive integer q. The supplied source [1] asserts $\mathcal{H}^m(\operatorname{Sing} V) = 0$, with no stability, minimization, orientation, or global multiplicity upper bound, and gives a round-sphere consequence through a cone construction.

That assertion is substantially stronger than smooth rectifiability: coverage almost everywhere by smooth graphs need not identify a whole neighborhood with one minimal graph of constant multiplicity. Nor can Allard's density-one theorem simply be applied after dividing by a tangent multiplicity Q: the necessary nearby density information is not supplied by that division. We do not resolve those issues anew.

In a flat comparison chart, with horizontal projection π, the source retains the signed mass and tilt measures

$$M = \pi_{\#}\mu - Q\, \mathrm{d}y, \quad D = \pi_{\#}(\operatorname{tr} P_{vv}\, \mu), \quad L = 2M - D, \quad \tau = |M| + D. \tag{1.1}$$

For a curved reference center, replace $\mathrm{d}y$ by its actual base volume, and add

$$\Lambda = \pi_{\#}\big([|z_{\mathrm{phys}}|^2 + |H_S|^2 + |\nabla^{\perp} H_S|^2]\mu\big), \qquad \tau = |M| + D + \Lambda.$$

The source then uses $L' = L - C_Q\Lambda$. Its domination $|L'| \leq C_0\tau$ is a measure inequality, not a sign assumption on M. Heights and curvature inside Λ retain physical units when a base chart is rescaled. In particular, a scaled height coordinate cannot silently replace z_{phys}.

1.1 Architecture of the assigned manuscript

The following are proof dependencies, not certifications of the complete chain. All nine scientific modules were literally read for this report.

1. `local.tex`: monotonicity, the full projected first variation, integer selection, configuration moments with $p > 2$, and Gaussian mass and height identities. The integer is selected on enlarged balls before using a total-variation estimate for M.

2. `signed-statement.tex`, `contact.tex`: a signed-excess target, a zero-mean heat potential, Brouwer contact selection, localization, and reduction of separated configurations to genuinely stationary lower-multiplicity restrictions.

3. `excess.tex`: future signed-mass envelope I, cumulative bank Z, clipped height moment, a low-frequency barrier and at most $Q - 1$ mergers. Later scales do not acquire an unproved pointwise contact sign.

4. `packing.tex`: normal-flux witnesses, a higher-moment controlled-density fraction, dyadic mass packing, and contact charging. Small base volume alone does not imply small projected varifold mass; the full defect M is retained.

5. `centering.tex`: minimal fitting, quadratic compatibility, separation after a failed fit, and a glued C^J center with a vanishing-density influence region.

6. `inward.tex`: a radial collar calculation retains the leading coefficient in $2A \le E + \text{errors}$; frequency estimates prove, rather than assume, inward doubling and eventual vanishing height.

7. `spheres.tex`: the stationary cone and slicing transfer the Euclidean claim to round spheres. This still depends on the Euclidean claim.

Our independent contributions concern the second item. Reading the whole source text is not an independent audit of every estimate or of its external regularity inputs.

2 A larger localization window

Fix a flat torus shape $X_1 = \mathbb{R}^m/\Gamma$ and let $X_R = RX_1$. All distances and Lipschitz constants use its flat physical metric, and $\mathrm{d}y$ is unnormalized volume. Put

$$G_s^R(x) = \sum_{\gamma \in R\Gamma} (2\pi s^2)^{-m/2} \exp(-|\widetilde{x} + \gamma|^2/(2s^2)), \qquad \lambda_s(x) = \int G_s^R(x-y)\,\mathrm{d}\lambda(y).$$

Thus $(d\,\mathrm{d}y)_s = d$, not $d/\operatorname{vol}(X_R)$. Constants below may depend on m, Γ, C_0, but not on η, d, R, T, τ. The fixed-shape qualification rules out collapsing tori with uncontrolled injectivity radius.

Lemma 2.1 (Tail with the retained Lipschitz factor)**.** *There are $c_*, C_*, s_* > 0$, depending on the fixed data, such that for $0 < s \le s_* R$, every $r > 0$, and any positive finite τ,*

$$\int_{\operatorname{dist}(x,y)\ge r} G_s^R(x-y)\frac{\operatorname{dist}(x,y)}{R}\,\mathrm{d}\tau(y) \le C_* \frac{\tau(X_R)}{R} s^{1-m} e^{-c_* r^2/s^2}. \tag{2.1}$$

Proof. For each lift $u = \widetilde{x-y} + \gamma$, $\operatorname{dist}(x,y) \le |u|$, and on the indicated set every lift has $|u| \ge r$. The elementary estimate

$$|u| e^{-|u|^2/(2s^2)} \le C s\, e^{-|u|^2/(4s^2)} \le C s\, e^{-r^2/(8s^2)} e^{-|u|^2/(8s^2)}$$

retains one factor s. For s/R bounded above, the final lattice Gaussian sum is uniformly bounded, by grouping lifts in lattice annuli of thickness comparable to R. This proves a pointwise bound on the kernel integrand and then (2.1). This argument also controls nonnearest periodic copies; no Euclidean tail is simply discarded. □

Theorem 2.2 (Improved source localization)**.** *Let $d > 0$, $0 < \eta < 1$, τ be a finite positive measure, $|L'| \le C_0\tau$, and $0 \le w \le 1$, $\operatorname{Lip} w \le R^{-1}$. Assume $T \ge 1$, $C_1 \ge 0$, and $\tau(X_R) \le dR^m e^{C_1 T}$. Define*

$$H = 1 + C_1 T + m\log(1/\eta), \qquad \rho = \frac{c\eta R}{\sqrt{H}}, \qquad f = wL' - \eta\nu, \quad \nu = \tau + d\,\mathrm{d}y. \tag{2.2}$$

For one sufficiently small fixed $c > 0$, every $x \in X_R$ and $0 < s \le \rho$ satisfy

$$|(wL')_s(x) - w(x)L_s'(x)| \le \frac{\eta}{4}\nu_s(x), \qquad f_s(x) \le w(x)L_s'(x) - \frac{3\eta}{4}\nu_s(x). \tag{2.3}$$

In particular the source's half-margin conclusion holds, with a quarter of the original margin available for an independently bounded error.

Proof. Set $r_* = \eta R/[4(1+C_0)]$. On $\operatorname{dist}(x,y) < r_*$, the commutator is at most $\eta\tau_s(x)/4$. On the complement, use $\operatorname{Lip} w \le R^{-1}$ and Lemma 2.1. After incorporating the fixed factor in r_*, this contribution is at most

$$Cde^{C_1 T} x^{1-m} e^{-c_2\eta^2/x^2}, \qquad x = s/R.$$

For $m > 1$, $x^{1-m}e^{-c_2\eta^2/x^2}$ is increasing up to $x^2 = 2c_2\eta^2/(m-1)$; for $m = 1$ it is increasing everywhere. Choose c so the entire asserted interval lies in that range and in the small-scale range of the lemma. At $x = c\eta/\sqrt{H}$, the last bound divided by ηd is

$$Cc^{1-m}H^{(m-1)/2}\exp\{H - 1 - c_2 H/c^2\}. \tag{2.4}$$

One can choose c so this is at most $1/4$ for all $H \ge 1$: first ensure $c_2/c^2 - 1 > (m-1)/2$, making the expression decreasing in H, and then ensure its value at $H = 1$ is at most $1/4$. The exponential in c^{-2} dominates c^{1-m}. Near and far contributions together give the first inequality in (2.3), and the definition of f gives the second. □

Remark 2.3 (What changed)**.** Source `contact.tex`, lines 149–209, proves its sufficient window $c\eta^2 R/T^2$ by a Gaussian split. We do not credit that method as new. The increment is retaining the distance factor in the far commutator and solving the resulting mass-versus-tail inequality at the larger scale (2.2). Even the coarser bound without that factor gives $m+1$ in place of m in H. For fixed η, the new window has $T^{-1/2}$, not T^{-2}; for fixed T and $\eta \downarrow 0$, it has order $\eta/\sqrt{\log(1/\eta)}$, not η^2. No best numerical c, varifold regularity rate, or measured computational speedup is asserted.

Corollary 2.4 (Exact contacts and a reserved error budget)**.** *Suppose the source's exact contact construction gives a positive measure h, of mass $f(X_R) > 0$, with*

$$(f-h)_{s_0}(x) \ge 0, \qquad \int_{s_0}^{s} v^2 (f-h)_v(x)\,\mathrm{d}\log v \ge 0.$$

If $s_0 \le s \le \rho$, then $w(x) > 0$ and

$$L'_{s_0}(x) \ge h_{s_0}(x) + \tfrac{3\eta}{4}\nu_{s_0}(x), \tag{2.5}$$

$$\int_{s_0}^{s} v^2 (L' - \tfrac{3\eta}{4}\nu)_v(x)\,\mathrm{d}\log v \ge 0. \tag{2.6}$$

More generally, a smoothed additive error e_s satisfying $|e_s| \le \eta\nu_s/4$ for every required x, s still leaves the half-margin bound $f_s + e_s \le w(x)L'_s - \eta\nu_s/2$.

Proof. If $w(x) = 0$, (2.3) is strictly negative because $d > 0$, contradicting $f_{s_0} \ge h_{s_0} \ge 0$. The initial inequality follows by division by $0 < w(x) \le 1$. After integration, (2.3) gives $w(x)\int v^2 L'_v \ge (3\eta/4)\int v^2\nu_v + \int v^2 h_v$; the right side is nonnegative. Division proves (2.6). The error assertion is immediate. It is conditional on a bound throughout the specified region, not on finitely many sampled values. □

2.1 An abstract sharpness boundary

Proposition 2.5 (Order obstruction with atomic data)**.** *On $X_1 = (\mathbb{R}/8\mathbb{Z})^m$, set $R = d = 1$, $w(y) = \min\{\operatorname{dist}(0,y), 1\}$, $y_\eta = 2\eta e_1$, $\tau = e^A\delta_{y_\eta}$, and $L' = \tau$, where $0 < \eta < 1/4$ and $A \ge 0$. Put*

$$H_0 = A + m\log(1/\eta), \qquad s = 2\eta/\sqrt{H_0}.$$

As $H_0 \to \infty$, the localized source has $f_s(0) > 0$, whereas the half-margin conclusion at $w(0) = 0$ would be negative. Consequently the order $\eta/\sqrt{A + m\log(1/\eta)}$ cannot in general be increased by an unbounded factor under measure domination and a mass bound alone.

Proof. The constant volume density is preserved by heat, so

$$f_s(0) = \eta\{e^A G_s^1(y_\eta) - 1\}.$$

The nearest Gaussian lift gives

$$\log(e^A G_s^1(y_\eta)) \geq \frac{H_0}{2} + \frac{m}{2}\log H_0 - m\log 2 - \frac{m}{2}\log(2\pi) \longrightarrow \infty.$$

Thus the conclusion fails at a fixed multiple of the asserted scale order. If positive total source mass is desired, choose $e^A > 8^m$; then $f(X_1) = \eta(e^A - 8^m) > 0$. Taking $A \to \infty$, or taking $\eta \to 0$ at any such fixed A, gives the two limiting regimes. These measures need not arise from stationary integral varifolds, so this is not a geometric counterexample or a sharpness claim for the full theorem. □

Example 2.6 (Source parameter comparison)**.** If $\eta = T^{-N}$ with fixed $N > 0$, our sufficient scale is comparable, up to fixed constants when $C_1 > 0$, to $RT^{-N-1/2}$. The source localization lemma supplies RT^{-2N-2}. But the separated-configuration recursion independently chooses $\rho/R = T^{-4N}$; our calculation alone does not relax that recursion. Likewise the outward construction chooses $\eta = e^{-2F_Q(T)}$ and $\rho = \eta^4 T^{-2}$. Changing its envelope, merger bands or packing rates would require checking those independent inequalities again.

3 Zero-mode heat potentials and their finite tails

For explicit Fourier constants, work on $X = (\mathbb{R}/2\pi\mathbb{Z})^m$, of volume $V_X = (2\pi)^m$. This is a change of stated normalization, not a change to probability volume. Let f be a finite real signed measure with mass $M_f > 0$. For any probability σ on a compact set of space–scale points, write

$$h = M_f(\pi_X)_\# \sigma, \qquad g = f - h, \qquad g(X) = 0, \qquad \|g\|_{\mathrm{TV}} \leq W := \|f\|_{\mathrm{TV}} + M_f.$$

Our total-variation norm is the full variation mass. Define

$$b(x,s) = \frac{1}{V_X} \sum_{k \in \mathbb{Z}^m \setminus \{0\}} \frac{e^{-s^2|k|^2/2}}{|k|^2} \widehat{g}(k) e^{ik\cdot x}, \qquad \widehat{g}(k) = \int e^{-ik\cdot y}\, \mathrm{d}g(y). \tag{3.1}$$

The bound $|\widehat{g}(k)| \leq W$ allows atoms. For every fixed $a > 0$, this series and all derivatives converge uniformly on compact scale intervals $s \geq a$, by Gaussian summability. It is the mean-zero inverse of $-\Delta$ followed by heat, and

$$\partial_{\log s} b = -s^2 g_s, \qquad \partial_t b = \Delta b, \quad t = s^2/2. \tag{3.2}$$

The mass-zero identity is exact. A pseudoinverse applied to incompatible nonzero-mass data would instead remove a constant source and would not give (3.2) for the original g.

Proposition 3.1 (Potential and source truncation are different)**.** *Let b_N and $g_{s,N}$ retain the modes $0 < |k|_\infty \leq N$, with $N \geq 0$. Put*

$$S_j = (2j+1)^m - (2j-1)^m, \qquad \mathcal{T}_r(a,N) = \sum_{j=N+1}^{\infty} S_j j^{-r} e^{-a^2 j^2/2} \quad (r = 0, 2).$$

Then uniformly for $x \in X$, $s \geq a$,

$$|b - b_N| \leq E_0 := \frac{W}{V_X}\mathcal{T}_2(a,N), \qquad |g_s - g_{s,N}| \leq E_1 := \frac{W}{V_X}\mathcal{T}_0(a,N). \tag{3.3}$$

For an explicitly finite upper bound define

$$q_N = \left(\frac{2N+5}{2N+3}\right)^{m-1} e^{-a^2(2N+3)/2}.$$

If $q_N < 1$, then for $r = 0, 2$,

$$\mathcal{T}_r(a, N) \le \frac{2m(2N+3)^{m-1}(N+1)^{-r}e^{-a^2(N+1)^2/2}}{1-q_N}. \tag{3.4}$$

Proof. The sup-norm shell $|k|_\infty = j$ has S_j points and $|k| \ge j$. Absolute summation gives (3.3). The mean-value formula applied to u^m gives $S_j \le 2m(2j+1)^{m-1}$. Consecutive terms $(2j+1)^{m-1}j^{-r}e^{-a^2j^2/2}$ have ratio at most q_N for $j \ge N+1$: both the polynomial ratio and Gaussian ratio decrease with j, while $(j/(j+1))^r \le 1$. Summing the geometric majorant proves the claim. □

The estimate is uniform over all such g, but only at a positive bottom. At fixed N, E_1 is not uniformly small as $a \downarrow 0$. This is a genuine analytic boundary, not just a defect in the geometric estimate: for an atomic mean-zero measure, $g_a(0)$ can grow like a^{-m}, whereas any fixed Fourier polynomial remains bounded. One must take $N = N(a) \to \infty$ when the bottom decreases and the desired error requires it.

3.1 A complete tail fixture

Take $m = 1$, $a^2 = 2\log 2$, and $N = 1$. Then $e^{-a^2j^2/2} = 2^{-j^2}$, so all shell calculations are rational,

$$q_1 = \frac{1}{32}, \qquad \mathcal{T}_0 \le \frac{4}{31}, \qquad \mathcal{T}_2 \le \frac{1}{31}.$$

For $W/V_X \le 1$, these bound the density error by $4/31$ and the potential error by $1/31$. Their difference matters: inserting a potential-tail bound where a source-tail bound is required would understate the uncertainty by a factor four even in this small example. This is a manufactured Fourier fixture, not a sampled varifold or a numerical contact for the source geometry.

4 A finite contact certificate with separate initial control

Let $\mathcal{D} \subset X \times [a, \infty)$ be closed and upward closed in scale, containing the full cylinder above a scale $b > a$. Let $K = \mathcal{D} \setminus \operatorname{int}\mathcal{D}$, where the interior is in $X \times (0, \infty)$, and suppose $K \subset X \times [a, b]$. This includes vertical sides and bottom points. Choose a finite h_K-net $z_i = (x_i, s_i)$ in K for the distance

$$d_K((x,s),(y,v)) = \operatorname{dist}_X(x,y) + |\log s - \log v|.$$

Let $p_i \ge 0$, $\sum p_i = 1$, define $\sigma = \sum_i p_i\delta_{z_i}$, and construct g, b, b_N from this same σ. It is not permissible to evaluate payoffs for one probability and report a different probability as the contact measure.

Theorem 4.1 (Finite payoff to cumulative contact)**.** *Suppose computed real numbers $\widetilde{b}_i$ satisfy $|\widetilde{b}_i - b_N(z_i)| \le \zeta$. Let*

$$\delta_N = \max_j \widetilde{b}_j - \min_{\{i:p_i>0\}} \widetilde{b}_i, \tag{4.1}$$

$$L_N = \frac{W}{V_X} \sum_{0<|k|_\infty \le N} (|k|^{-1} + b^2)e^{-a^2|k|^2/2}, \tag{4.2}$$

$$\epsilon = \delta_N + 2\zeta + L_N h_K + 2E_0. \tag{4.3}$$

For every i with $p_i > 0$ and every $s \geq s_i$,

$$\int_{s_i}^{s} v^2 g_v(x_i)\, \mathrm{d}\log v = b(z_i) - b(x_i, s) \geq -\epsilon. \tag{4.4}$$

The theorem makes no assertion that $g_{s_i}(x_i) \geq 0$.

Proof. Spatial differentiation of b_N bounds its gradient by the $|k|^{-1}$ part of (4.2). Logarithmic scale differentiation multiplies each potential coefficient by $-s^2|k|^2$, and $a \leq s \leq b$ gives the other part. Integrating along a shortest torus path and then a logarithmic scale interval proves the stated Lipschitz bound on the full slab. Every point of K is therefore bounded above in payoff by $\max_j b_N(z_j) + L_N h_K$. The arithmetic errors in the maximum and selected minimum cost 2ζ. The two potential tails cost $2E_0$, giving

$$b(z_i) \geq \max_K b - \epsilon.$$

The heat maximum principle gives $\sup_{\mathcal{D}} b \leq \max_K b$. For completeness, truncate at any upper scale above b and apply the maximum argument to $b - \xi t$. An interior maximum contradicts $(\partial_t - \Delta)(b - \xi t) = -\xi$; a maximum at the artificial top has nonnegative one-sided time derivative and nonpositive spatial Laplacian, giving the same contradiction. Points outside K have an open neighborhood in $\mathcal{D}$, and upward closure supplies the needed preceding cylinder at its artificial top. Let $\xi \downarrow 0$ and increase that top. Finally $(x_i, s) \in \mathcal{D}$ by upward closure, and integration of (3.2) proves (4.4). □

Proposition 4.2 (Two legitimate initial controls)**.** *At a selected node, a computed density u_i with $|u_i - g_{s_i,N}(x_i)| \leq \zeta_1$ gives*

$$g_{s_i}(x_i) \geq u_i - \zeta_1 - E_1. \tag{4.5}$$

Alternatively assume $|\partial^2_{\log s} b(x_i, s)| \leq B$ for $s_i \leq s \leq e^{\ell} s_i$, with $0 < \ell < \infty$. Under Theorem 4.1, every $0 < t \leq \ell$ gives

$$g_{s_i}(x_i) \geq -\frac{1}{s_i^2}\left(\frac{\epsilon}{t} + \frac{Bt}{2}\right). \tag{4.6}$$

If $B > 0$ and $\sqrt{2\epsilon/B} \leq \ell$, the right side is $-\sqrt{2B\epsilon}/s_i^2$.

Proof. The first statement is (3.3) and the triangle inequality. For the second, put $F(t) = b(x_i, s_i e^t)$. The cumulative certificate gives $F(t) - F(0) \leq \epsilon$. Taylor's integral formula and $|F''| \leq B$ give $F(t) - F(0) \geq tF'(0) - Bt^2/2$. Since $F'(0) = -s_i^2 g_{s_i}(x_i)$, rearrangement proves (4.6); minimization gives its last assertion. □

One valid, generally coarse, B on a slab $[a, b_1]$ is

$$B = \frac{W}{V_X} \sum_{k \neq 0} (2b_1^2 + b_1^4|k|^2) e^{-a^2|k|^2/2}.$$

This follows by differentiating (3.1) twice. The series is finite for $a > 0$, but the bound is not uniform as $a \downarrow 0$. A fixed payoff error divided by an arbitrarily short logarithmic increment is not an initial derivative certificate.

Example 4.3 (A payoff gap does not give a derivative sign)**.** At one outward vertical in the circle let

$$b(0, s) = -\epsilon e^{-k^2(s^2 - a^2)/2}, \qquad k \geq 1.$$

This is the value at $x = 0$ of a mean-zero heat potential proportional to $-\cos(kx)$. It stays within ϵ above its initial value for all $s \geq a$, yet $g_a(0) = -k^2\epsilon < 0$. This is a negative control for an inference from a vertical payoff gap, not a counterexample to exact contact existence. Its increasing curvature as k grows explains why the additional B in Proposition 4.2 is necessary.

Corollary 4.4 (How finite errors enter the localized source)**.** *Use Theorem 2.2 and a finite contact certificate on scales up to ρ. Suppose $g_{s_i}(x_i) \geq -e_i$ with $e_i < (3\eta/4)\nu_{s_i}(x_i)$. Then $w(x_i) > 0$ and*

$$w(x_i)L'_{s_i}(x_i) \geq h_{s_i}(x_i) + (3\eta/4)\nu_{s_i}(x_i) - e_i.$$

For $s_i \leq s \leq \rho$, one also has

$$\int_{s_i}^{s} v^2(L' - \tfrac{\eta}{2}\nu)_v(x_i)\,\mathrm{d}\log v \geq \frac{\frac{\eta}{4}\int_{s_i}^{s} v^2\nu_v(x_i)\,\mathrm{d}\log v - \epsilon}{w(x_i)}. \tag{4.7}$$

Thus nonnegative cumulative half-margin requires, for the specified interval, that its retained quarter-margin exceed ϵ. It does not automatically hold on every sufficiently short interval.

Proof. If $w = 0$, the localized upper bound on f, positivity of h, and the strict hypothesis on e_i contradict $f - h \geq -e_i$. The initial assertion follows without changing the sign of any term. Integration gives $w\int v^2L'_v \geq (3\eta/4)\int v^2\nu_v - \epsilon$, after dropping the nonnegative integral of h. Subtract $w(\eta/2)\int v^2\nu_v$ and use $w \leq 1$. Division by the now positive w proves (4.7). □

5 Selection geometry, smoothing, and what is not certified

Proposition 5.1 (A single-scale variational formulation)**.** *For $K = X \times \{a\}$, let*

$$K_a(x) = \frac{1}{V_X}\sum_{k\neq 0}\frac{e^{-a^2|k|^2/2}}{|k|^2}e^{ik\cdot x}, \qquad \mathcal{J}(\sigma) = \int (K_a * f)\,\mathrm{d}\sigma - \frac{M_f}{2}\iint K_a(x-y)\,\mathrm{d}\sigma(x)\,\mathrm{d}\sigma(y).$$

On probabilities on X, $\mathcal{J}$ is continuous and concave. A probability maximizes it if and only if it is carried by the maximizers of its own payoff $b_\sigma(\cdot, a)$.

Proof. For a finite real signed measure ξ,

$$\iint K_a(x-y)\,\mathrm{d}\xi(x)\,\mathrm{d}\xi(y) = \frac{1}{V_X}\sum_{k\neq 0}\frac{e^{-a^2|k|^2/2}}{|k|^2}|\widehat{\xi}(k)|^2 \geq 0.$$

Absolute convergence justifies the identity. Compactness of probabilities and continuity give existence. For another probability ν, the right derivative of $\mathcal{J}((1-t)\sigma+t\nu)$ at zero is $\int b_\sigma\,\mathrm{d}(\nu-\sigma)$. At a maximum it is nonpositive for every ν, and using point masses says $\int b_\sigma\,\mathrm{d}\sigma = \max_X b_\sigma$. Continuity then gives the support statement. Conversely that support statement makes every such derivative nonpositive; the exact quadratic expansion has an additional nonpositive term, so it implies maximality. □

This is classical quadratic-energy/KKT reasoning, not a new general optimization theorem. It clarifies a boundary of the source mechanism. For nodes at varying scales, the interaction is

$$A_{ij} = K_{s_i}(x_i - x_j),$$

which is generally not symmetric. For two nodes with the same spatial location and different scales, $K_{s_1}(0) \neq K_{s_2}(0)$, because each nonzero Fourier term strictly decreases with scale. Thus the off-diagonal entries already differ. The source's Brouwer-softmax existence proof works without symmetry; it neither proves that Picard iteration converges nor supplies a complexity bound. A finite probability passing (4.1) is useful irrespective of how it was found, but successful finite solves do not certify the decreasing-bottom limit used in the source proof.

Proposition 5.2 (Weak smoothing bias, not reconstruction)**.** *For any finite signed λ and Lipschitz test φ on X,*

$$\left| \int \varphi \, \mathrm{d}(G_s * \lambda - \lambda) \right| \leq \mathrm{Lip}(\varphi)\, s\sqrt{m}\, \|\lambda\|_{\mathrm{TV}}.$$

There is no corresponding total-variation convergence for all measures: $\|G_s * \delta_0 - \delta_0\|_{\mathrm{TV}} = 2$ for every $s > 0$.

Proof. The periodized Gaussian is the law of sZ modulo the lattice, where Z is standard Gaussian in $\mathbb{R}^m$. Torus displacement is at most $s|Z|$; integration against $|\lambda|$, followed by $\mathbb{E}|Z| \leq \sqrt{m}$, proves the bound. The heat image of a point mass has a smooth density and is mutually singular with that point mass. The variation mass of their difference is therefore two. □

Propagation (3.2), Fourier truncation (3.3), and this weak approximation concern three different errors. None implies regularity of the unsmoothed varifold.

6 Ram comparison, consequences, and limitations

6.1 The original private proof connection

The freshly read original TR8 proof [6], lines 81–177, starts with an actual weak elliptic realization, proves the coercive inverse on its named spaces, separates Neumann compatibility from the choice of constant gauge, and proves Green equivalence only after retaining the boundary data. Lines 196–236 construct the heat semigroup spectrally, justify the energy solution and Duhamel formula, and distinguish its infinite spatial support from a tail truncation. Those are the relevant proof mechanisms here. We do not transfer its Dirichlet realization unchanged onto a torus: in (3.1) the domain is periodic and the zero eigenmode is removed only because $g(X) = 0$. Also our input can be an atomic measure rather than the L^2 data of that TR8 energy theorem; positive-bottom absolute Fourier convergence is proved separately.

The supplied core-pat-5 document [7], freshly read at lines 88–134 and 189–300, is a secondary organizing comparison, not a replacement for those original PDE proofs. Its actual metric formula, compatibility proof, induced-norm inverse-tail proof and spectral-frame calculation explain why a finite representation needs its physical integration weights and a discarded-space error. Our Fourier coefficients use physical volume V_X and complex conjugation in the quadratic form for precisely that reason. Neither an exact finite matrix inverse nor a frame on retained modes proves a continuum contact theorem. The beginning of its unused discarded-space proposition at lines 296–300 was read, but its continuation was not; no proof claim is based on it.

These are connections to established operator arguments in the local corpus, not historical novelty claims for heat kernels, Fourier series, Gaussian splitting, the maximum principle, or KKT conditions. Earlier project reports and parent checks are not counted as original proof evidence or as our own tests.

6.2 Advantages and costs

The larger localization radius gives a concrete relaxation of one source inequality without changing the signed physical measures. Keeping $\nu = \tau + d\,\mathrm{d}y$, rather than just τ, pays commutators where local mass is tiny. The atomic obstruction identifies why a mass-sensitive logarithmic term cannot simply be omitted. The finite contact certificate is independently checkable once its true integration errors, net coverage, total variation and cutoff are bounded. It distinguishes a failure to meet a sufficient certificate from nonexistence of a contact.

The costs remain substantial. A Fourier box has $(2N+1)^m - 1$ modes. Positive-bottom tail bounds worsen as the bottom shrinks; net size depends on the actual entrance boundary.

The constants in the torus estimates depend on its shape and on C_0, which in the application contains dimension and multiplicity data. No efficient method for constructing K, integrating an arbitrary varifold, or finding a small-gap probability has been proved here. Strict verification for general transcendental coefficients would require validated arithmetic, not ordinary floating-point sampling.

6.3 Precise downstream boundary

To substitute our exact localization theorem in the source, the same extension-by-zero domination, full-torus constant density and negative source margin must be preserved. The exact contact theorem can then use the larger window. But the later source proof also requires Gaussian chart-cutoff errors and curved commutators to fit the chosen margin, envelope initialization, lower-integer stationary restrictions, height configuration moments, finite mergers, witness packing and the inward center construction. None of those follows from (2.2).

An approximate finite contact has further obligations. Its initial defect must pass (4.5) or another independently proved derivative estimate. Its cumulative error must be smaller than the interval's bank allowance in (4.7); a fixed $\epsilon > 0$ does not work on every arbitrarily short interval. Letting the bottom tend to zero requires a compatible sequence of cutoffs, meshes and errors, with a compactness argument. We have not proved that the actual source geometry supplies an effective such sequence, and we do not replace its exact Brouwer existence argument. Ordinary smoothing alone changes neither the unresolved geometric dependencies nor the quantifier *almost every support point*.

6.4 Feasible next steps

One precise extension is to retain spatially varying certified commutator budgets instead of a common $\eta\nu/4$. This would require checking their integrated bank payment on every outward interval, not just their total mass. A second is a validated positive-bottom solver whose finite gap, quadrature remainder and Fourier complement are jointly bounded on a supplied entrance boundary. The varying-scale nonsymmetry must be retained in any convergence proof. Finally, any attempt to improve the source's iterated-log signed-excess rate should redo its node-scale, merger and packing inequalities; the present larger window alone does not give that conclusion.

7 Current primary literature and evidence

7.1 Public status as checked on October 9, 2026

The publisher page for Brena–Decio–De Lellis [2] identifies volume 36 (2025), pages 409–443, and a publication date of February 3, 2026. Its abstract describes height bounds, Lipschitz approximation and excess decay. The arXiv abstract and version record for Brena–De Lellis–Franceschini [3] identify v1 of March 1, 2025 and a smooth-rectifiability conclusion. Kolasiński's September 17, 2026 v1 [4] likewise states rectifiability of every finite order and smooth rectifiability. These are not the same statement as neighborhood regularity of all stationary integral varifolds.

The September 17, 2026 v1 abstract of De Lellis–Hirsch–Lihn–Spolaor [5] concerns a smooth Riemannian metric with a multiplicity-two planar tangent and infinite local topology at an exceptional point, for $m \geq 3$. Its setting and conclusion must not be presented as either a Euclidean almost-everywhere counterexample or verification of the supplied source. The online checks here are of these primary abstracts and version or publisher metadata, not fresh full audits of their proofs. The source repository paper page was also located publicly; the cached-text byte comparison uses the captured October trees, not a claim about a freshly downloaded live branch.

7.2 Literal scope and reproducibility

The complete assigned public text inventory comprises 15 files and 6,620 literal lines, including nine scientific modules, wrapper, bibliography, two figure sources, Unicode map and README. All were displayed successfully in bounded chunks. Two private originals were prehashed and 297 lines were freshly displayed; their exact unread complements total 1,694 lines. All other private files and external proof bodies remain outside this read claim. The PRE receipt was persisted before the first scientific display. After a connection interruption, the completed four-section read intervals were transcribed from retained execution context; the transcription is not backdated. All subsequent intervals are separately recorded, and all PRE/POST hashes agree. Heading searches and truncated discovery output are excluded from literal-read claims.

The owned script `M663-checks.py` executes 435 exact cases and 10,918 exact assertions, plus 27 float64 consistency cases with one assertion each. Exact cases use rational heat factors 2^{-hk^2}, quarter-turn Fourier phases, shell counts, positive quadratic forms, the integrated heat identity, and signed error allocation. They include varying-scale nonsymmetry and the failure of an approximate payoff gap to force an initial nonnegative source. The 27 numerical cases compare two evaluations of the atomic logarithmic formula; they are not interval certificates of an inequality. The script and its result receipt are hash-linked. None of the parent's independent cases, earlier reports' cases, or foreign checkers is included in those totals.

8 Conclusion

At the abstract measure interface actually used by the source, retaining the Lipschitz factor enlarges the sufficient Gaussian localization window and leaves a quantified margin. The same scale order is forced by explicit atomic data under those abstract assumptions. Positive-bottom spectral tails then support a finite contact certificate with separate cumulative and initial error controls. These are notable local deductions; their value is precision about usable radii and certification boundaries, not a new claimed solution of the global regularity problem. C3 is assessed for those deductions independently of the catalogue's L3 applicability.

References

[1] OpenAI, *Almost-everywhere regularity of stationary integral varifolds*, September 23, 2026. Pinned public manuscript. Assigned source modules were read in full; the global theorem is attributed, not independently certified here.

[2] C. Brena, S. Decio and C. De Lellis, *Remarks and conjectures on stationary varifolds dedicated to Enrico Bombieri*, Rend. Lincei Mat. Appl. 36 (2025), 409–443. Publisher abstract and metadata; DOI 10.4171/RLM/1077.

[3] C. Brena, C. De Lellis and F. Franceschini, *C^∞ rectifiability of stationary varifolds*, arXiv:2503.00649v1, March 1, 2025. Abstract and version record checked; proof not independently audited.

[4] S. Kolasiński, *Rectifiability of every finite order for stationary integral varifolds*, arXiv:2609.19979v1, September 17, 2026. Abstract and version record checked.

[5] C. De Lellis, J. Hirsch, Z. Lihn and L. Spolaor, *Stationary varifolds with singularities II*, arXiv:2609.20647v1, September 17, 2026. Abstract and version record checked.

[6] M. Subbarao, confidential original TR8, local directory `RamNonlin-IE-PDE-astra/TR8-lin-non-lin`, file `TR8-lin-non-lin.tex`. Fresh literal scope: lines 81–177 and 196–236. Weak inverse, gauge and heat-equivalence proof bodies; no upload.

[7] M. Subbarao, confidential core-pat-5, local directory `RamPatsCore50-TRs/TR-core-pat-5`, file `TR-core-pat-5.tex`. Fresh literal scope: lines 88–134 and 189–300. Secondary metric/inverse/spectral comparison; not a substitute for the original PDE proof or a whole-corpus audit.